TOPIC 04

Sums of Squares, R-squared and OLS

Splitting the variability of the data into explained and unexplained parts, measuring the quality of a fit with R^2, and how least squares finds the line

WHAT YOU WILL LEARN

- Define SST, SSR and SSE and state SST = SSR + SSE
- Compute and interpret R^2 = SSR/SST
- Judge whether an R^2 is high or low in context
- Explain what ordinary least squares (OLS) minimises
- Compute b_1 and b_0 from the data by formula
- Name other ways of fitting a regression line

NOTE **Where this connects**

Topic 03 drew the regression line $\hat{y} = b_0 + b_1x$ and said it is the 'best-fitting' line. This topic says what *best* means and how good the fit is. It uses the **ANOVA framework**: ANOVA stands for analysis of variance, the idea of splitting total variation into sources, and the lecture applies it here. We break the total variation in y into the part the line explains and the part it does not. Variance and standard deviation come from descriptive statistics.

1. Three sums of squares

Take any observed y_i, its mean $\bar{y}$ and its fitted value $\hat{y}_i$. Three vertical distances matter. Each is squared and added over all n observations.

Name	Formula	What it measures
SST (total)	$\Sigma(y_i - \bar{y})^2$	**Total variability** of the data around its mean
SSR (regression)	$\Sigma(\hat{y}_i - \bar{y})^2$	**Explained** variability: how far the line moves away from the mean. It shows how well the line fits the data
SSE (error)	$\Sigma(y_i - \hat{y}_i)^2 = \Sigma e_i^2$	**Unexplained** variability: the squared residuals. We want it as small as possible

$$SST = SSR + SSE$$ total variability = explained + unexplained

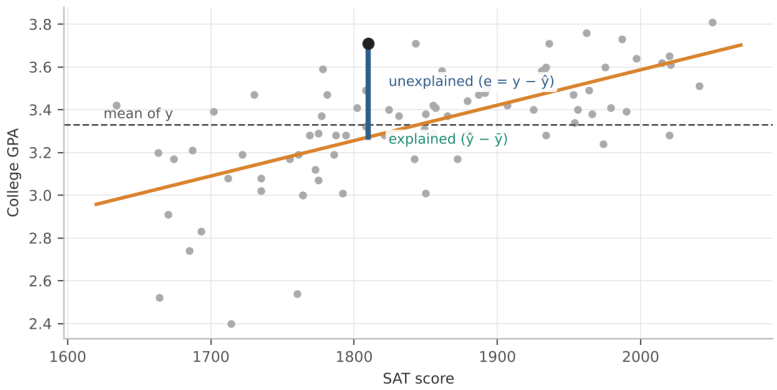


Figure 1. For one student the total gap $y - \bar{y}$ splits into an explained part (mean to line) and an unexplained part (line to point). Squaring and summing over all students gives SSR and SSE.

Why the pieces add up

SST = SSR + SSE is not true for any line; it holds for the OLS line because the residuals add to zero and are uncorrelated with the fitted values, so the cross-term $2\Sigma(\hat{y} - \bar{y})(y - \hat{y})$ vanishes.

Five points by hand. x = 1, 2, 3, 4, 5 and y = 2, 4, 5, 4, 5 (the example of Topic 01). OLS gives $b_1 = 6/10 = 0.6$ and $b_0 = 4 - 0.6 \times 3 = 2.2$.

x	y	$\hat{y} = 2.2 + 0.6x$	$(y - \bar{y})^2$	$(\hat{y} - \bar{y})^2$	$(y - \hat{y})^2$
1	2	2.8	4	1.44	0.64
2	4	3.4	0	0.36	0.36
3	5	4.0	1	0.00	1.00
4	4	4.6	0	0.36	0.36
5	5	5.2	1	1.44	0.04
Sum			SST = 6	SSR = 3.6	SSE = 2.4

SSR + SSE = 3.6 + 2.4 = 6 = SST, and $R^2 = 0.60$, which equals $r^2 = 0.775^2 = 0.60$ from Topic 01.

2. A worked example with the SAT and GPA data

Quantity	Value (84 students)	Check
SST	6.1234	Total variation of GPA around its mean
SSR	2.4861	Explained by the line
SSE	3.6373	Left unexplained
SSR + SSE	6.1234	Equals SST

The lecture workbook for the adjusted R^2 topic reports the same total, SST = 6.1234, for this data set, which is a useful cross-check on our calculation.

3. R-squared

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

R^2 is the **share of the total variability that the regression explains**. It is a relative measure between 0 and 1.

- **$R^2 = 0$:** the regression line explains none of the variability; $\hat{y}$ is just $\bar{y}$.

- **$R^2 = 1$:** the line explains all of the variability; every point sits on the line.
- In simple regression $R^2 = r^2$, the square of the correlation (Topic 02).

For our data $R^2 = 2.4861/6.1234 =$ **0.4060**, and $r^2 = 0.4060$. The lecture rounds this to 0.406, or about 41%: SAT scores explain about 41% of the variability of college GPA in this sample.

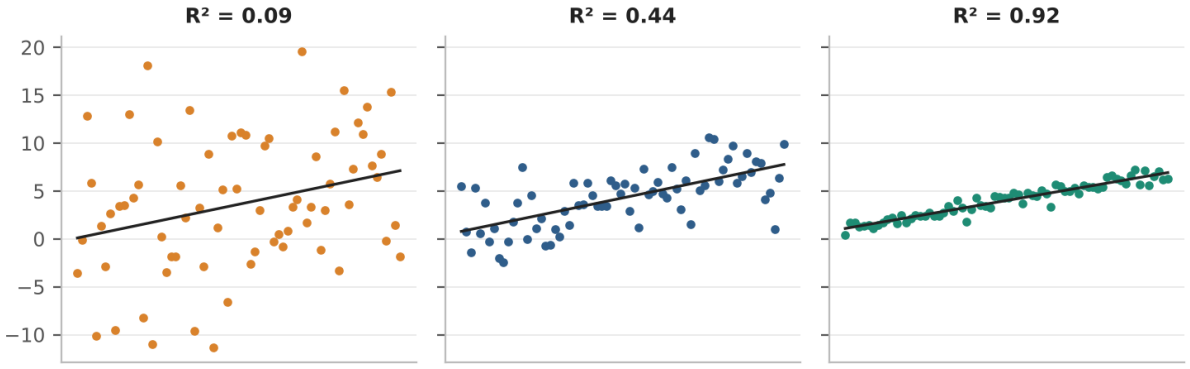


Figure 2. Simulated data with the same underlying line but increasing noise from right to left. The closer the points hug the line, the higher R^2.

Is 41% good?

The lecture's guidance is that R^2 is neither good nor bad by itself; it depends on the subject.

Field / outcome	Typical R^2	Why
Physical laws, engineering measurements	0.9 or higher	Few hidden causes, precise measurement
Behaviour, income, grades, health	Often 0.2 to 0.5	Many causes outside the model

Income, for example, depends on household income (including parents and spouse), education, years of experience, country, language spoken and more. A model with a few of these can still explain well under half of it. An R^2 of 0.2 can be useful there. In our case, 0.41 is far from 0.9, so SAT is clearly **not the only thing** that determines college GPA; we are missing something (Topic 06 adds more variables).

> **COMMON MISTAKE** **High R^2 is not proof of a good model**
> R^2 never decreases when you add more variables, even useless ones. And a high R^2 says nothing about causation. Topic 07 introduces the adjusted R^2, which penalises needless variables.

4. Ordinary least squares (OLS)

In practice you let software do the arithmetic. Beginners often use Excel, SPSS, SAS or Stata; professionals commonly use R or Python.

Many lines could be drawn through a cloud of points. **OLS** picks the one that makes the **sum of squared errors smallest**.

> **DEFINITION** **OLS principle**
> Choose b_0 and b_1 to minimise $SSE = \Sigma(y_i - b_0 - b_1 x_i)^2$. Lower error means better explanatory power, so the OLS line is the one closest to all the points. Squaring makes every error positive, so over- and under-predictions cannot cancel out, and it punishes large errors more.

Calculus (setting the derivatives of SSE to zero) gives closed-form answers:

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{s_{xy}}{s_x^2} \qquad b_0 = \bar{y} - b_1\bar{x}$$

With our data: $\Sigma(x - \bar{x})(y - \bar{y}) = 1501.56$ and $\Sigma(x - \bar{x})^2 = 906912.70$, so $b_1 = 0.001656$, and $b_0 = 3.3302 - 0.0016557 \times 1845.27 = 0.2750$. You could do this on paper, but with hundreds or thousands of rows regression is always done by software.

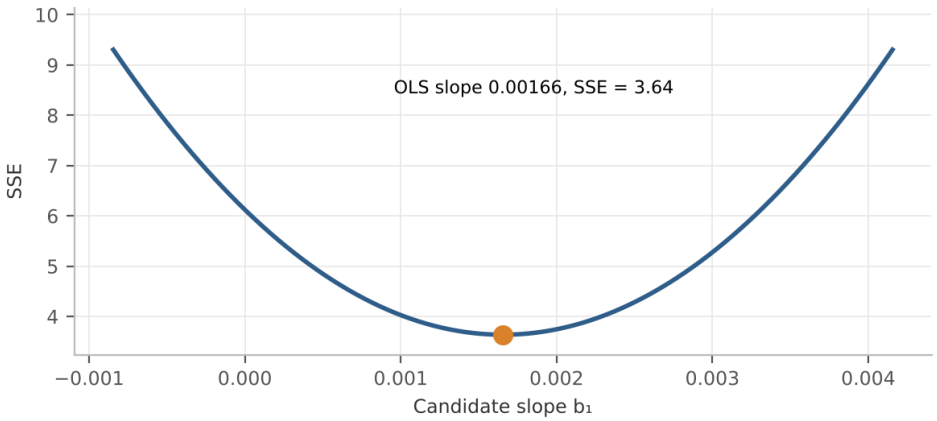


Figure 3. For lines through the point of means, SSE as a function of the slope is a parabola with its minimum at the OLS slope. Every other slope leaves more error.

Other ways to find a regression line

OLS is the standard method for linear regression. Others exist, such as **Bayesian regression**, **kernel regression** and **Gaussian process regression**. They are used for special situations and are not needed in this course.

5. In Python

```
import numpy as np

data = np.loadtxt("satgpa.csv", delimiter=",", skiprows=1)
x, y = data[:, 0], data[:, 1]

b1 = np.sum((x - x.mean()) * (y - y.mean())) / np.sum((x - x.mean()) ** 2)
b0 = y.mean() - b1 * x.mean()
yhat = b0 + b1 * x

sst = np.sum((y - y.mean()) ** 2)
ssr = np.sum((yhat - y.mean()) ** 2)
sse = np.sum((y - yhat) ** 2)
print(round(sst, 4), round(ssr + sse, 4))     # equal
print(round(ssr / sst, 4))                    # R-squared
```

SUMMARY

- SST = SSR + SSE: total variability = explained + unexplained.
- R^2 = SSR/SST ranges from 0 to 1; in simple regression it equals r^2.
- What counts as a good R^2 depends on the field; behavioural outcomes often sit at 0.2 to 0.5.
- OLS chooses b_0 and b_1 to minimise SSE; the formulas are $b_1 = s_{xy}/s_x^2$ and $b_0 = \bar{y} - b_1\bar{x}$.
- Alternatives to OLS exist but are beyond this course.

Practice questions

Try each question before reading the answer underneath it.

Q1. SST = 50 and SSE = 20. Find SSR and R^2.

Answer. SSR = 50 − 20 = 30 and R^2 = 30/50 = 0.60.

Q2. A model has R^2 = 0.85 and another has 0.25. Does the first always mean a better model?

Answer. Not necessarily. Whether 0.25 or 0.85 is good depends on the subject, and R^2 also rises just by adding variables. Compare using the adjusted R^2 and the context.

Q3. Why does OLS minimise squared errors instead of the plain sum of errors?

Answer. Plain errors cancel (positives with negatives, so the sum is zero for many lines); squares are all positive and penalise large misses.

Q4. $\bar{x} = 10$, $\bar{y} = 20$, $b_1 = 2$. Find b_0.

Answer. $b_0 = 20 - 2 \times 10 = 0$. The line passes through the origin.

Q5. What does $R^2 = 0$ tell you about the best-fitting line?

Answer. It is flat ($b_1 = 0$): $\hat{y} = \bar{y}$ for every x, so x explains none of the variability.